A robust interface for the reusable operator

Question. Can contrast-normalizing successor rows before sparse expansion preserve the frozen parity readout under continued symbol corruption, while retaining V193's clean label-free adaptation?

Starting evidence

After 512 noisy post-change blocks in V193, the frozen-readout operator scored 70.31% on changed pairs and 83.85% on still-valid pairs. The jointly updating reference reached 92.71% and 96.88%. Within the noisy suite these arms saw the same structural observations and used the same map updates. The clean frozen operator scored 99.48% and 100%. Clean and noisy V193 suites used different worlds.

The saved observed-transition graph comparator scored 100% on both panels at blocks 128 and 512 in each suite. Its changed-pair score was 0% at blocks 8 and 32. Its rolling window mixes pre-change and post-change observations at early checkpoints. Its balanced two-group prior makes it a task reference. The late gap shows the corrupted observations can support recovery under that prior.

Intervention

Keep direct online TD, the one-step and two-step pathways, the fixed sparse expansion, and the local readout. For each pathway m, maintain a vector S equal to the sum of its successor rows over the fixed byte vocabulary. When TD changes one row, add that row's delta to S. Set the common component mu = S / 12. Initialize S from the actual map at initialization or resume. The support is the same declared vocabulary as V193, with no partition labels or hidden cycle order.

For each operand x, compute v = M(x) minus mu, then use v / (||v|| + epsilon), with proposed epsilon = 10^{-8}. Concatenate the transformed rows in the original operand and pathway order. Keep projection, tanh, top-k selection and post-selection normalization unchanged. Apply the transform during initial readout teaching and all evaluation. Freeze the trained readout and newline table at the graph change.

Independent corruption can attenuate the parity contrast while retaining a common component. Centering and scaling test whether the readout depends too strongly on that balance. The finite-sample TD maps and sparse nonlinear features can also distort the signal. No part of the transform uses an answer or modifies a TD update.

Paired arms and controlled exposure

Main arms

Raw frozen. V193 successor rows with a frozen post-change readout. **Contrast-normalized frozen.** Center and scale rows before projection, with the same readout freeze. **Centered-only frozen.** Subtract the centroid without scaling to test the contribution of rescaling. **Raw joint.** V193 raw rows with post-change readout feedback, to measure recovery with labels.

Train each interface's readout from the same initialization and pre-change teaching schedule, with the same projection and active count. All arms start the change phase with identical maps. Readout weights may differ across interfaces because they were trained on different features. Clone each pre-change state into clean and noisy continuations. Hold observations, query order, feedback opportunities and map-update budgets paired. Only raw joint receives post-change answer labels.

Worlds, streams, and probes

Use 12 fresh paired worlds, each with clean and 10% independently corrupted post-change continuations, for 96 lifetimes. Preserve V193's clean warm-up, pre-change noise phase, adjacent opposite-parity swap, fixed byte parser and learned newline table. Use the same latent blocks and query order in both continuations. Apply the noise mask only to background observations. Supply neither phase nor corruption flags to the learner.

Seal separate changed and still-valid panels of 16 ordered pairs each, balanced by answer label, with reverse pairs withheld throughout. Score first generated answer plus generated newline before feedback, at blocks 0, 8, 32, 128 and 512. Also report answer-only accuracy and obsolete responses. At block 512 the frozen arms have 8,192 post-change background bytes, 14,336 TD row updates and zero new relation labels. Record actual counts.

References and diagnostics

Run the V193 rolling transition-count comparator from the observed stream with pre-change polarity feedback only. Include the constant and exact or unordered pair baselines. Record per-pathway centroid and contrast norms, sparse active-set overlap, readout margins and per-world trajectories. Use matched clean/noisy forks for drift comparisons and record the centroid's memory and per-update overhead.

Keep the fixed query parser and background channel explicit. The task measures reuse of a previously taught parity relation over untaught pairs. Exact-pair recitation cannot retrieve held pairs, and the two-step structure and fixed operand roles remain strong task priors.

Decision and execution gates

Proposed primary decision at block 512

Noisy changed pairs. Contrast normalization must achieve at least 85% joint accuracy and beat raw frozen by at least 10 percentage points. The paired 95% interval for the gain must exclude zero.

Noisy still-valid pairs. Require at least 90% joint accuracy and no more than a three-point mean loss against raw frozen. **Clean guardrail.** Require at least 95% on both panels and no more than a two-point mean loss against raw frozen on either. Report paired intervals and per-world results for every comparison. These are proposed thresholds to freeze before final runs.

Require the raw frozen clean reference to reach at least 95% on both panels before interpreting the result. Report whether the noisy raw reference exhibits enough headroom for the prespecified gain. If it does not, classify the test as uninformative for promotion and keep the gate unchanged. Treat block 128 and the full curve as latency and diagnostic outcomes.

Interpretation

Pass. Use the normalized interface as the reference for this noise regime. If centered-only gives comparable results, attribute the supported change to removing the common component and favor the simpler interface pending confirmation.

Gain with clean loss. Keep the raw reference. Inspect whether the transform discards useful contrast magnitude. **No gain with a strong graph reference.** Retire this candidate on the tested configuration. Inspect distortions from sparse feature selection and readout compatibility before proposing another interface. If the graph reference also degrades, examine observation quality before attributing loss to the readout.

Three engineering cycles before the official roster

1. Verify incremental centroids against batch recomputation, finite features near zero contrast, exact raw-path parity with V193, and map equality across arms for identical streams.

2. Run complete engineering lifetimes. Check raw-reference headroom, balanced reverse-closed holdouts, freeze hashes, zero-label isolation, causal scoring and exposure receipts.

3. Verify deterministic checkpoint resume in a fresh process, rejection of corrupt receipts, and serial-to-parallel equivalence. Seal source, world roster, thresholds and reduction code before official runs.

Separate work

V192 context separation addresses conflicting sources with contextual evidence. Its scientific run remains unresolved after a budget admission stop. Keep that study and factual-memory integration separate from V194. Basis: V193 saved results at ca74be1f and the user-provided audit, including V192 at 7fb2ff17.